Can a Protective Interphase Coating Extend Lithium-ion Battery Cycle Life?

Can a protective interphase coating extend lithium-ion battery cycle life?

Prepared by AI co-scientist on 2026-08-04. For research purposes only.

Draft for internal review. This report holds proposed hypotheses and the reviews they received, not verified findings; every claim must be independently verified against primary sources before it is acted upon.

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Research Goal Details

Goal

Can a protective interphase coating extend lithium-ion battery cycle life?

Requirements

1. Must include uncoated control cells
2. Must define specific independent variables (coating material, thickness) and dependent variables (capacity retention, Coulombic efficiency)
3. Must specify sample size and randomization of cell testing channels
4. Must operate within safe voltage and temperature limits to prevent thermal runaway

Attributes

- Research mode proposed: experimental — no experiment was performed in producing this report
- Intended claim: hypothesis
- Analysis scope: the full requested analysis, carried out as desk work
- Approval profile: auto
- Assumption: The coating can be uniformly applied without degrading baseline battery performance
- Assumption: Cycle life degradation in uncoated batteries is primarily driven by interphase instability
- Assumption: The testing environment accurately simulates real-world battery usage conditions
- Correctness: scored one to five by the evidence and correctness review
- Novelty: scored one to five by the novelty review
- Feasibility: scored one to five by the methods and feasibility review
- Impact: scored one to five by the impact review
- Safety: scored one to five by the safety and governance review

Criteria

Success criteria — what would make the goal met:

- Demonstrate a statistically significant increase in cycle life (e.g., >10% improvement in cycles to 80% capacity retention) for coated cells vs. controls
- Maintain or improve initial Coulombic efficiency
- Replicate findings across a minimum of 3 independent batches of cells

Comparison criteria — what every idea was scored against:

- Evidence support (alignment with verified literature and known degradation mechanisms).
- Feasibility (compatibility with standard roll-to-roll manufacturing and cell assembly).
- Information gain (ability to isolate the specific mechanism of degradation and protection).
- Impact (potential percentage increase in cycle life and capacity retention).
- Cost/time (duration of cycling tests, synthesis cost, and equipment requirements).
- Reproducibility (variance across the 3 required independent batches).
- Safety (risk of thermal runaway, venting, or lithium plating during testing).

Research Overview

Top ideas

Ranked Research Ideas

Prepared by AI co-scientist on 2026-08-04. For research purposes only.

Warning: this overview was assembled mechanically from what each stage of the run recorded. Every sentence in it restates one of those records, and no model was asked to read the run as a whole, so where two stages disagree the disagreement is reported rather than resolved.

1. Research Goal

This report addresses a single research goal, stated under Goal on the cover and restated below in the form the run executed. The goal was declared as experimental work with an intended claim of hypothesis, which fixes the standard of proof every idea below is held to. Nothing in this document is a finding; the ideas are proposals that have been reviewed, ranked and stress-tested against each other, and each still requires independent verification before it is acted upon. A source satisfies an evidence gate only when its original content has been inspected and mapped to the exact claim. Auto approval is a workflow convenience and never constitutes scientific, safety, ethics, or institutional approval.

Scoping restated that goal as the question the run went on to execute: Does the application of a specific protective interphase coating on the electrodes extend the cycle life of lithium-ion batteries compared to uncoated controls under standardized cycling conditions? This is the narrower formulation, and it is the one the ideas below were generated, reviewed and ranked against. Where it and the goal on the cover differ, an idea that answers this question has not necessarily answered the goal.

Warning: eight stage gates in this run were approved automatically under the auto approval profile, covering scoping the goal, literature discovery, idea generation, independent review, tournament ranking, evolution of the shortlist, clustering by mechanism, and meta-review. Nobody read what those stages produced before it was accepted. An acceptance recorded here means only that the work was complete and well-formed, not that a person agreed with it.

The goal carries four explicit constraints that bound what may legitimately be proposed. They are set out and numbered under Requirements on the cover, and those numbers are what the rest of this report refers to. No stage of this run screened the ideas against them mechanically, and compliance was not a condition of being ranked or promoted. What the record does hold is 35 reviews of the ideas, printed in full under each. Read against the constraints, constraints one and four are reached by a reviewer's own words: constraint one, in one review under Feasibility; and constraint four, in two reviews under Safety. None of those reviewers was asked to check a constraint and none records a verdict on one, so this says where to look rather than whether an idea complies. Nothing any reviewer wrote reaches constraints two and three, so on those this report shows no coverage. The match is on wording, so a reviewer who addressed a constraint in other terms than the goal states it in would not be counted here.

The plan also records three assumptions the work rests on, listed under Attributes on the cover. They matter because an idea inherits every assumption its goal was framed with. Each is an inference rather than an observation, none was tested in this run, and a reviewer who disagrees with one should expect the corresponding ideas to weaken.

The safety and governance review recorded a fatal flaw against one hypothesis in this run, and it was answered by a named human rather than by the workflow. One hypothesis was withdrawn from the population and did not compete. Every decision is reprinted in full, with its flaw and its written justification, under Governance adjudications below; a safety decision summarised is a safety decision a reader cannot check. The other reviews recorded fatal flaws against a further seven ideas, which nobody adjudicated: those stand as written, under the ideas that carry them.

No experiment was performed in this run, no dataset was accessed and no measurement was taken. Every stage of it is desk work: a literature search, a set of proposals written by models, and reviews of those proposals by other models. So every quantitative statement below is an expectation recorded by whichever specialist proposed it, and the research mode named on the cover, experimental, describes the work being proposed rather than any work that was done.

Stopping criteria were declared up front so the run could not drift. Thermal runaway, venting, or cell rupture occurs during testing; initial capacity of coated cells is >15% lower than uncoated controls; and coating process proves incompatible with standard cell assembly. These are the conditions under which further work would have added cost without adding decision value, or must not continue at all. Nothing in this workflow enforces them; they bind whoever takes the work forward.

2. Evaluation Criteria

Every idea in this report was assessed against the seven comparison criteria set out on the cover, settled before the ideas were written. The tournament did not read them: every match in it was decided by arithmetic on the review scores rather than by a judge, so the criteria bear on the ranking below only through whatever the reviewers made of them.

Five independent reviews were run against each idea: evidence and correctness, novelty, methods and feasibility, impact, and safety and governance. Safety is read separately from impact because it decides whether the work may proceed at all rather than whether it is worth proceeding. Each review answers one question on a five-point scale. The recommendation sets the number — advance is five, revise it first is three, evidence too thin to judge on is two, reject it is one — and the reviewer's own stated confidence then moves it one point up at 0.80 or above and one point down below 0.30. A printed four is therefore a confidently held revise and a printed two a confidently held rejection, so the number carries the verdict and the conviction behind it together and neither can be read off it alone. A reviewer who records a fatal flaw caps the score at two whatever the recommendation, because an unresolved fatal flaw is disqualifying rather than merely costly.

Success for the goal as a whole was defined separately from per-idea scoring, and is stated under Criteria on the cover, in its own block apart from the comparison criteria the scoring used. No stage of this run scored an idea against it. An idea can therefore score well on the comparison criteria and still fail to advance the goal, and no ranking below closes that gap: it has to be closed by reading the success criteria against whichever idea is being considered.

3. Main Research Directions

Warning: the clustering by mechanism in this section is a fixed template, not the specialist's own reasoning. The specialist's answer came back incomplete or malformed and was replaced, so the content below states what the workflow requires rather than what a model concluded. Treat it as a placeholder and re-run the stage before relying on anything in this section.

Warning: the evidence gate for this run was waived, so the material below was admitted without meeting the verification standard the goal declared. The waiver is recorded against a researcher working through the command line. Waiving the gate is a distinct act from accepting the stage's output, and whichever approval regime this run used applies to the second of those, not the first. The statements in this section are exploratory leads and are not verified findings; nothing here should be cited as established, and the gate should be re-run before any idea grounded on it is acted upon.

Discovery identified the directions along which the literature is currently moving, and these framed what the generator was asked to produce: three of them, listed under Research directions below. Each is a region of the problem rather than a proposal in itself.

A 5 nm Al_2O_3 coating optimally enhanced cycling performance on LFP cathodes, retaining 67% of its capacity after 100 cycles at a high 1C rate compared to 57% for the uncoated sample [1].

When coated with 1.0 wt% WO_3, the NCM-811 cathode exhibited a high fifth-cycle discharge capacity of 167.8 mAh/g and an outstanding capacity retention of 87.1% over 100 cycles [2].

Applying a carbon coating thicker than 15 nm resulted in remarkably poor rate performance, as the excessive thickness completely paralyzed ion transport [3]. Discovery recorded this finding as arguing against the hypothesis the question puts, not for it.

Triethyl phosphite (TEPi) fails to yield a protective positive electrode film and cannot block the generation of reactive oxygen radical species, resulting in explicitly inferior electrochemical performance [4]. Discovery recorded this finding as arguing against the hypothesis the question puts, not for it.

Discovery returned six further findings, which carry the same standing as those above and are stated here so that nothing the report cites is missing from its references.

A CuTF coating on the separator maintained 95% capacity after 280 cycles compared to 83% for the uncoated control, effectively increasing cycle life by a factor of three [5]. Discovery recorded two qualifications on it: Specific to CuTF coating on PE separators, and results may vary with different coating materials.

Interphase layers that are less permeable to Li^+ ions restrict electrolyte flow, increase overall cell impedance, and lead to power fade rather than extending useful life [6]. Discovery recorded this finding as arguing against the hypothesis the question puts, not for it. Discovery recorded two qualifications on it: Refers generally to SEI layers rather than engineered artificial coatings, and depends on the permeability of the specific layer.

A hybrid coating layer reduced permeability, causing DC internal resistance to increase and discharge capacity to decrease at high rates (>1.0 C), while showing no difference in discharge capacity at low rates compared to reference cells [7]. Discovery recorded this finding as arguing against the hypothesis the question puts, not for it. Discovery recorded two qualifications on it: Observed specifically at high discharge rates (>1.0 C), and specific to the tested hybrid organic/inorganic layer.

The pilot-scale synthesis process for protective coatings was repeated three times to determine reproducibility, yielding inter-sample molecular weight differences of less than 1.1% [7]. Discovery recorded one qualification on it: Demonstrates reproducibility of the coating synthesis, not necessarily the long-term cycling variance across independent batches.

Protective coatings are evaluated using symmetrical and asymmetrical cells to assess stability and Coulombic efficiency against bare, uncoated control cells under standardized constant current rates [8]. Discovery recorded one qualification on it: Describes methodological setup rather than universal outcomes.

Lithium-ion testing must operate within strict safety voltage and temperature ranges, utilizing Battery Management Systems (BMS) to prevent thermal runaway, overcharge, and localized overheating [9]. Discovery recorded one qualification on it: General safety governance requirement for all Li-ion testing, not specific to coated cells.

Mapping the generated ideas back onto the problem produced four clusters. The clustering stage recorded no mechanism that tells them apart, so the names below group the ideas without saying what the grouping rests on. Evidence First holds two ideas. Mechanism First holds two ideas. Analogy Transfer holds two ideas. Competing Explanation holds one idea. Clustering is meant to show where funding two ideas buys one idea's worth of information, and that reading is not available here: a shared name is not a shared failure mode.

4. Candidate Ideas

The generator produced eight ideas across the strategies described above, each stated as a claim that can be shown to be wrong. They are set out here in rank order, each with the same five-row grid so that they can be read against one another. The grid carries the idea in the specialist's own words: the mechanism it rests on, the prediction that separates it from its neighbours, the test that would show it wrong, the dependency it turns on, and its principal risk. The prose above each grid does not repeat those; the reviews each idea received and the matches it played are in its own section later in this report.

Each idea below carries a grounding verdict. What the recurring ones mean is the same wherever they appear, so it is set out here rather than under every idea that has one. Four of them are marked uncited. They cite no evidence anywhere in this report. Nothing here says whether evidence for them exists — only that none was retrieved and none was checked — so each stands on its reasoning, and the reviews under it are the only assessment of it. Three of them are marked unverified. Every claim they cite exists, but none has been verified against its source, so each idea rests on retrieved text rather than on checked evidence.

One of those ideas is not ranked below. The >10% Extension in Cycle Life was withdrawn after the safety and governance review recorded a fatal flaw, so it never entered the tournament. It is kept in the numbering below, unranked, with the decision that removed it and the reason given.

4.1 Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating

Applying a 5 nm Copper-Tricarboxylate (CuTF) metal-organic framework coating on the electrode extends cycle life by >10% to 80% capacity retention compared to uncoated controls at 1C discharge rates.

Beyond the prediction in the grid below, it is separated from its competitors by two further predictions: initial Coulombic efficiency of coated cells will be equal to or greater than controls, and internal resistance growth over 100 cycles will be significantly lower in coated cells. It has two competing readings to displace: the CuTF coating increases initial impedance, leading to lower initial capacity and failing the >15% capacity drop stopping criterion; and the coating degrades electrochemically at high voltages, providing no long-term benefit. It finished rank 1 on an Elo of 1262. Its grounding is marked unverified.

Category	Description
Mechanism	Evidence indicates that CuTF coatings can maintain 95% capacity after 280 cycles. An ultrathin MOF layer prevents transition metal crossover and acid attack while its porous structure maintains ionic pathways, avoiding the impedance trade-off seen in thicker coatings.
Discriminating prediction	Coated cells will reach 80% capacity retention at least 10% later than uncoated controls.
Falsifier	Cycle n=15 coated and n=15 uncoated control cells (randomized across cycler channels) at 1C within safe voltage limits (3.0-4.2V). Falsified if the mean cycles to 80% capacity retention for coated cells is not statistically significantly >10% higher than controls across 3 independent batches.
Key dependency	Ability to uniformly synthesize and apply 5 nm CuTF coatings.

Category	Description
Principal risk	Coating process may be incompatible with standard cell assembly.

4.2 A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode

A sacrificial lithium-carbonate rich composite pre-coating on the anode extends cycle life by >10% by preferentially reacting with trace water and HF to form a stable SEI, analogous to a sacrificial zinc anode in marine corrosion.

Beyond the prediction in the grid below, it is separated from its competitors by two further predictions: first-cycle irreversible capacity loss may be higher for coated cells, but subsequent Coulombic efficiency will be superior; and gas generation during the first cycle will be higher for coated cells. It has two competing readings to displace: the sacrificial reaction generates excessive gas, causing cell rupture or delamination of the electrode; and the resulting SEI is highly resistive, leading to power fade. It finished rank 2 on an Elo of 1246. Its grounding is marked uncited.

Category	Description
Mechanism	In corrosion engineering, sacrificial materials protect the primary substrate. A reactive pre-coating can consume electrolyte impurities during the first formation cycle, creating a highly stable SEI that prevents long-term degradation of the uncoated baseline.
Discriminating prediction	Coated cells will show >10% improvement in cycles to 80% capacity retention.
Falsifier	Cycle n=15 coated and n=15 uncoated cells. Falsified if the cycle life to 80% retention does not improve by >10%, or if the initial capacity of coated cells is >15% lower than uncoated controls due to excessive sacrificial lithium consumption.
Key dependency	Controlled environment (dry room) for applying the moisture-sensitive sacrificial coating.
Principal risk	Excessive gas generation during the sacrificial reaction could lead to cell venting or rupture.

4.3 A 2 nm Atomic Layer Deposition (ALD) Al2O3 Coating

A 2 nm Atomic Layer Deposition (ALD) Al2O3 coating on NCM cathodes extends cycle life by >10% by chemically scavenging HF generated from LiPF6 degradation, preventing transition metal dissolution.

Beyond the prediction in the grid below, it is separated from its competitors by two further predictions: post-mortem analysis of the electrolyte will show significantly lower concentrations of dissolved Ni, Co, and Mn in coated cells; and initial Coulombic efficiency will be maintained or improved. It has two competing readings to displace: the Al2O3 coating acts purely as a physical barrier rather than a chemical scavenger, and the 2 nm coating is too thin to provide meaningful HF scavenging over hundreds of cycles. It finished rank 3 on an Elo of 1199. Its grounding is marked unverified.

Category	Description
Mechanism	Cycle life degradation in high-nickel cathodes is primarily driven by HF attack from electrolyte decomposition. Al2O3 reacts with HF to form stable AlF3, protecting the active material without blocking Li+ transport due to its ultrathin nature.
Discriminating prediction	Coated cells will demonstrate >10% improvement in cycles to 80% capacity retention.
Falsifier	Cycle n=15 coated and n=15 uncoated cells across 3 independent batches. Measure dissolved transition metals in the electrolyte post-mortem via ICP-MS. Falsified if cycle life is not extended by >10% OR if transition metal dissolution is not significantly reduced in coated cells.

Category	Description
Key dependency	Access to ALD equipment capable of conformal 2 nm deposition.
Principal risk	ALD precursors may react with the cathode surface, altering baseline performance.

4.4 A 5 nm Boron Nitride (BN) Coating

A 5 nm Boron Nitride (BN) coating extends cycle life by >10% primarily by acting as a thermal heat spreader that eliminates localized hot spots on the electrode surface, rather than by acting as a chemical barrier against the electrolyte.

Beyond the prediction in the grid below, it is separated from its competitors by two further predictions: in-situ thermal imaging will show a significantly more uniform temperature distribution across the BN-coated electrode compared to controls, and cells coated with a thermally insulating material of the same thickness will not show the cycle life improvement. It has two competing readings to displace: the BN coating improves cycle life purely through chemical passivation, independent of its thermal conductivity; and localized hot spots are not a primary driver of degradation under standardized cycling conditions. It finished rank 4 on an Elo of 1184. Its grounding is marked uncited.

Category	Description
Mechanism	Localized Joule heating accelerates SEI degradation and transition metal dissolution. A thermally conductive but electrically insulating coating homogenizes the temperature profile, slowing down these thermally activated degradation mechanisms.
Discriminating prediction	BN-coated cells will show >10% improvement in cycles to 80% capacity retention.
Falsifier	Cycle n=15 BN-coated cells, n=15 uncoated cells, and n=15 cells coated with a thermally insulating polymer of the same thickness. Falsified if BN-coated cells do not outperform both other groups by >10% in cycle life, or if thermal imaging shows no reduction in localized hot spots.
Key dependency	Capability to deposit uniform 5 nm BN coatings and a thermally insulating control coating.
Principal risk	Modified cell designs for thermal imaging may not accurately represent standard commercial cell behavior.

4.5 A 15 nm Hybrid Organic-inorganic Coating Extends Cycle Life

A 15 nm hybrid organic-inorganic coating extends cycle life by >10% at low discharge rates (0.2C) but decreases cycle life compared to uncoated controls at high rates (2.0C) due to restricted Li+ permeability.

Beyond the prediction in the grid below, it is separated from its competitors by two further predictions: at 2.0C, coated cells will reach 80% capacity retention faster than uncoated controls; and DC internal resistance will be significantly higher for coated cells at 2.0C. It has two competing readings to displace: the coating provides universal protection, extending cycle life at both 0.2C and 2.0C; and the coating is too thick and paralyzes ion transport completely, failing at all C-rates. It finished rank 5 on an Elo of 1172. Its grounding is marked unverified.

Category	Description
Mechanism	Evidence shows hybrid coatings reduce permeability, increasing DC internal resistance at high rates (>1.0C) while showing no capacity difference at low rates. This suggests a rate-dependent impedance trade-off where the coating acts as an electrical bottleneck under high current.
Discriminating prediction	At 0.2C, coated cells will show >10% improvement in cycles to 80% capacity retention vs controls.

Category	Description
Falsifier	Test n=15 coated and n=15 uncoated cells at 0.2C, and another set of n=15 coated and n=15 uncoated cells at 2.0C. Falsified if coated cells do not show >10% life extension at 0.2C OR if they do not show worse capacity retention than controls at 2.0C.
Key dependency	Precise control over hybrid coating thickness to exactly 15 nm.
Principal risk	High-rate cycling of high-impedance cells may lead to localized overheating and thermal runaway.

4.6 A 10 nm Ductile PEDOT:PSS Conductive Polymer Coating

A 10 nm ductile PEDOT:PSS conductive polymer coating on silicon-graphite anodes extends cycle life by >10% by accommodating volume expansion and preventing mechanical pulverization of the SEI.

Beyond the prediction in the grid below, it is separated from its competitors by two further predictions: cross-sectional SEM will show significantly less particle cracking and SEI thickness growth in coated cells compared to controls, and coulombic efficiency will stabilize faster in coated cells. It has two competing readings to displace: the polymer coating degrades electrochemically at low anode potentials, and the coating restricts lithium diffusion, causing lithium plating and rapid capacity fade. It finished rank 6 on an Elo of 1169. Its grounding is marked uncited.

Category	Description
Mechanism	Cycle life degradation in high-capacity anodes is driven by mechanical stress and continuous SEI fracture during lithiation/delithiation. A ductile, conductive coating maintains electrical contact and stabilizes the interphase during volume changes.
Discriminating prediction	Coated cells will achieve >10% improvement in cycles to 80% capacity retention.
Falsifier	Cycle n=15 coated and n=15 uncoated cells. Falsified if coated cells do not achieve >10% improvement in cycles to 80% capacity retention, or if post-mortem cross-sectional SEM shows equal or greater particle pulverization compared to controls.
Key dependency	Synthesis of battery-grade, moisture-free PEDOT:PSS.
Principal risk	Residual water in the polymer coating could react with the electrolyte, causing severe gas generation and cell venting.

4.7 A Self-assembling Amphiphilic Block-copolymer Coating on the Cathode

A self-assembling amphiphilic block-copolymer coating on the cathode extends cycle life by >10% by acting as a selective filter that allows bare Li+ transport while sterically hindering the co-intercalation of large solvent molecules, analogous to biological ion channels.

Beyond the prediction in the grid below, it is separated from its competitors by two further predictions: electrolyte consumption over 200 cycles will be significantly lower in coated cells, and initial Coulombic efficiency will be improved due to reduced solvent oxidation. It has two competing readings to displace: the polymer pores become blocked by degradation products, paralyzing ion transport; and the polymer dissolves in the organic carbonate electrolyte over time. It finished rank 7 on an Elo of 1168. Its grounding is marked uncited.

Category	Description
Mechanism	Biological cell membranes use size and charge exclusion to achieve selective permeability. An engineered polymer with specific pore sizes can prevent solvent molecules (e.g., EC, DEC) from reaching the active material surface, halting parasitic side reactions.
Discriminating prediction	Coated cells will show >10% improvement in cycles to 80% capacity retention.

Category	Description
Falsifier	Cycle n=15 coated and n=15 uncoated cells. Falsified if coated cells fail to show >10% cycle life improvement, or if impedance spectroscopy shows the coating completely paralyzes Li^+ transport resulting in an initial capacity >15% lower than controls.
Key dependency	Synthesis of block-copolymers with precise sub-nanometer pore sizes.
Principal risk	Polymer dissolution could increase electrolyte viscosity and decrease overall cell safety.

4.8 The >10% Extension in Cycle Life (withdrawn)

Withdrawn after review. J. Reviewer (battery safety officer) withdrew this hypothesis from the population in answer to a fatal flaw recorded by the safety and governance review. It did not enter the tournament, so it carries no rank and no Elo. The flaw, reprinted verbatim: "Annealing PVDF-containing electrodes at 400°C will cause binder decomposition, releasing highly toxic and corrosive hydrogen fluoride (HF) gas, posing an extreme chemical safety hazard to researchers and equipment." The reason J. Reviewer (battery safety officer) gave for the decision is reprinted in full under Governance adjudications below.

The hypothesis as written was: The >10% extension in cycle life observed in coated cathodes is caused by the high-temperature annealing step (e.g., 400°C) used during the coating process, which heals surface microcracks, rather than the physical presence of the coating material itself. It is reproduced here so that the decision to withdraw it can be judged against what was actually proposed. This slot is the whole of its presence in the report; there is no section of its own further down.

5. Comparison of Candidate Ideas

Warning: the tournament ranking in this section is a fixed template, not the specialist's own reasoning. The specialist's answer came back incomplete or malformed and was replaced, so the content below states what the workflow requires rather than what a model concluded. Treat it as a placeholder and re-run the stage before relying on anything in this section.

The ideas were compared head to head over twelve matches, every one of which produced a winner. Ratings started level and moved only on the outcome of a match, so an idea's final rating is a statement about who it beat rather than about how it read in isolation.

Every match was decided by an arithmetic score comparison rather than by argument. No exchange underlies any individual result, so the ordering should be read as a summary of the reviews rather than as an independent judgement of the ideas.

Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating finished first on four wins against no losses, ending on an Elo of 1262. Its nearest rival, A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode, ended 16 points behind. No single match in this tournament moved a rating by more than 16 points — the K factor of 32 is the most one could ever move it, and only against an idea the ratings had already written off — so that gap is inside what one different result could have produced, and the two should be read as level rather than ordered. It was never paired against A 10 nm Ductile PEDOT:PSS Conductive Polymer Coating and A Self-assembling Amphiphilic Block-copolymer Coating on the Cathode, so two ideas of the six it is ranked above never met it: those positions rest on shared opponents rather than on a result between them.

The full standings follow. Ideas that finished level on rating share a position; the order among them below is the sort's and not a result.

Position	Idea	Elo	Record (W-L)
1	Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating	1262	4-0
2	A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode	1246	4-1
3	A 2 nm Atomic Layer Deposition (ALD) Al2O3 Coating	1199	2-2
4	A 5 nm Boron Nitride (BN) Coating	1184	1-2
5	A 15 nm Hybrid Organic-inorganic Coating Extends Cycle Life	1172	1-3
6	A 10 nm Ductile PEDOT:PSS Conductive Polymer Coating	1169	0-2
7	A Self-assembling Amphiphilic Block-copolymer Coating on the Cathode	1168	0-2

The ranking did not converge under the rule this run applies: two consecutive rounds ending on the same top four, and a final round that moves no rating by more than five per cent of the 1200 every idea starts on. This tournament did not record how far its final round moved the ratings, so the second half of that rule was not tested here and the ordering below rests on the match results alone. No two consecutive rounds ended on the same top four, which is where this run fell short of the first half of that rule.

The shortlist carried forward into evolution and re-review was Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating; A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode; A 2 nm Atomic Layer Deposition (ALD) Al2O3 Coating; and A 5 nm Boron Nitride (BN) Coating. Shortlisting is a budget decision rather than a verdict: an idea left out may still be correct, and one carried forward may still fail its first go/no-go test.

A 2 nm Atomic Layer Deposition (ALD) Al2O3 Coating was shortlisted while carrying a reviewer's recommendation to reject it. The shortlist is drawn on tournament rating and does not read the recommendations, so the two are not in conflict by any rule this run applied — but nothing reconciled them either, and the rejection stands unanswered against an idea the run went on to develop.

6. Comparative Analysis by Review Criterion

Ranking compresses five separate judgements into one number, so this section unpacks them again. An idea that leads overall while trailing on correctness is a different proposition from one that leads on correctness while trailing on safety, and the two call for different next steps. The ordering in section five was computed from these scores, no judge having been available to compare the ideas, so a criterion that separates the field here is one that moved the ranking there.

On correctness, the ideas averaged 2.0 out of five. Every review on this criterion came in at 2, so it separates nothing and cannot be used to justify a choice between the ideas.

On novelty, the ideas averaged 3.7 out of five. Four of the ideas tied highest at 5, and A 2 nm Atomic Layer Deposition (ALD) Al2O3 Coating; A 15 nm Hybrid Organic-inorganic Coating Extends Cycle Life; and A 10 nm Ductile PEDOT:PSS Conductive Polymer Coating tied lowest at 2. That is a spread of three points with four ideas at the top of it, wide enough to separate the field.

On feasibility, the ideas averaged 2.7 out of five. Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating scored highest at 5, and five of the ideas tied lowest at 2. That is a spread of three points with one idea at the top of it, wide enough to separate the field.

On impact, the ideas averaged 4.6 out of five. Four of the ideas tied highest at 5, and A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode; A 5 nm Boron Nitride (BN) Coating; and A

Self-assembling Amphiphilic Block-copolymer Coating on the Cathode tied lowest at 4. The spread is narrow enough that this criterion did not separate the field and should not be used to justify a choice between them.

On safety, the ideas averaged 4.6 out of five. Four of the ideas tied highest at 5, and A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode; A 2 nm Atomic Layer Deposition (ALD) Al2O3 Coating; and A 5 nm Boron Nitride (BN) Coating tied lowest at 4. The spread is narrow enough that this criterion did not separate the field and should not be used to justify a choice between them.

One objection recurred across the field, which is a stronger signal than any single score. The specific expected outcomes (>10% improvement) are speculative without evidence — raised against at least four of the seven ideas (quoted with one idea's figures). Three of the ideas escaped it, so it is not a property of the goal: whether a choice inherits it depends on which idea is chosen.

7. Comparison with Existing Solutions

An idea is only worth pursuing if it is not already settled, so each one was reviewed against what the field is understood to do today. That comparison is only as good as the prior art the workflow could actually see, and its limits are stated below rather than left implicit.

The comparison is against the literature this run retrieved: the ten findings stated in full under Main Research Directions above. The standing of those findings is the standing of every novelty judgement below.

The novelty reviews judged the following to add something beyond current practice: Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating; A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode; A 5 nm Boron Nitride (BN) Coating; and A Self-assembling Amphiphilic Block-copolymer Coating on the Cathode. These are the novelty reviewers' own judgements, made against the retrieved literature listed above rather than against a systematic prior-art search, and each review is printed in full under its idea. The verdict is not unqualified: the same review recorded an objection against every one of them. A four or a five is the reviewer's answer to how much is new, not a record that nothing was raised, and what was raised is printed under the idea it was raised against.

The following were judged too close to established practice, or too poorly evidenced to place: A 2 nm Atomic Layer Deposition (ALD) Al2O3 Coating; A 15 nm Hybrid Organic-inorganic Coating Extends Cycle Life; and A 10 nm Ductile PEDOT:PSS Conductive Polymer Coating. An idea in this group is not necessarily wrong; it is unlikely to return information the field does not already hold.

8. Key Findings and Unexpected Connections

Warning: the clustering by mechanism, evolution of the shortlist, and meta-review in this section are fixed templates, not the specialists' own reasoning. Their answers came back incomplete or malformed and were replaced, so the content below states what the workflow requires rather than what a model concluded. Treat it as a placeholder and re-run the stages before relying on anything in this section.

This run established nothing about the world. What follows is the state of the idea space it produced: which proposal leads, which reviews went against the field, and where the meta-review's account of the round and the reviews themselves disagree.

The sources this report draws on are listed under References. Every one of them is a lead rather than a verified reference: the workflow recorded where a statement came from, but inspecting the original and confirming that it says what is attributed to it remains outstanding.

Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating is the strongest proposal this run produced: it finished rank 1 on an Elo of 1262 with four wins from four matches, and its claim, predictions and falsifier are set out under its own heading in Top ideas in detail, below. What it would risk, in full: coating process may be incompatible with standard cell assembly, and potential for thermal runaway if the MOF catalyzes electrolyte decomposition.

Fifteen reviews closed at two or below, the band that covers a rejection, a hesitant finding that the evidence is too thin to judge on, and any review at all that recorded a fatal flaw. No idea in the run escaped the band. All of them except Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating drew more than one such review, which is a judgement several reviewers reached separately rather than one reviewer's reservation.

The meta-review excluded every ranked idea from any recommendation, stating that an unresolved fatal flaw stands against each. Exclusion here is procedural and not reversible by a better score elsewhere. The reviews carry the flaw in every case: one was recorded against each idea named here.

The meta-review also listed The >10% Extension in Cycle Life as excluded, which it was, but not by the meta-review: it was withdrawn from the population by a named person, and the adjudication is set out with the flaw and the reason given under Governance adjudications below. That is a human decision on the record, not a summary-stage one.

9. Recommendations and Next Steps

Warning: the meta-review in this section is a fixed template, not the specialist's own reasoning. The specialist's answer came back incomplete or malformed and was replaced, so the content below states what the workflow requires rather than what a model concluded. Treat it as a placeholder and re-run the stage before relying on anything in this section.

The recommendation below is a proposal for what to do next, conditional on the verification steps named with it. It is not an approval, and no part of this workflow can supply one.

No idea cleared the bar for an unconditional recommendation. Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating is the least weak of the set and is the sensible place to spend the next increment of effort, but it should be funded as an evidence-gathering exercise rather than as a test of the idea itself. That is the standing of the whole set rather than a clearance for this one: a reviewer recorded a fatal flaw against it, and the meta-review excluded it from any recommendation, and closing that comes before any of the work below.

The decision is sensitive to a short list of specific evidence. A safety finding that resolves the fatal flaw recorded against the leading idea, which is what currently keeps every candidate off the recommendation. The outcome of the falsifying test the proposal names: cycle n=15 coated and n=15 uncoated control cells (randomized across cycler channels) at 1C within safe voltage limits (3.0-4.2V). Falsified if the mean cycles to 80% capacity retention for coated cells is not statistically significantly >10% higher than controls across 3 independent batches. A direct measurement of the prediction that separates it from the field: coated cells will reach 80% capacity retention at least 10% later than uncoated controls. Evidence for the competing reading its ranking assumes away, which is that the CuTF coating increases initial impedance, leading to lower initial capacity and failing the >15% capacity drop stopping criterion. Obtaining any one of these is worth more than another round of generation, because it would move the ranking rather than lengthen it.

The immediate next step for the leading idea is the go/no-go work its specialist set down in advance. Verify initial capacity of coated cells is not >15% lower than uncoated controls, and confirm coating thickness is 5 nm ±

1 nm via TEM before full cell assembly. Running that, once the flaw and the exclusion above are closed and the governance obligations below are discharged, is what converts this report from a set of proposals into a decision.

Before any physical, clinical or data-access step, the governance obligations attached to the goal have to be discharged by a named owner: adherence to laboratory safety protocols for handling lithium-ion batteries and reactive materials, proper disposal procedures for hazardous battery waste, and calibration of battery cyclers and environmental chambers. None of these can be discharged by the workflow itself, and an automatic approval recorded in it satisfies none of them.

Research directions

- Deploy Hybrid Deposition Architectures.
- Prioritize Ductile Interphases for Solid-State Systems.
- Optimize Dual-Coating Strategies.

Review summary

- Correctness: mean 2.0 of five across seven reviews, range 2 to 2.
- Novelty: mean 3.7 of five across seven reviews, range 2 to 5.
- Feasibility: mean 2.7 of five across seven reviews, range 2 to 5.
- Impact: mean 4.6 of five across seven reviews, range 4 to 5.
- Safety: mean 4.6 of five across seven reviews, range 4 to 5.

Governance adjudications

One governance adjudication is recorded for this run: a person's answer to a fatal flaw the safety and governance review raised. It is set out below with the flaw it responds to and the reason given, both reprinted word for word. The wording is theirs; nothing here is a summary, because a reader has to be able to judge the decision rather than learn that one was taken.

No other fatal finding was left unanswered in this run.

The names below are as entered by whoever ran the adjudication. This system does not authenticate them, so a name here is a record of what was claimed at the time and carries only as much weight as the process that produced it.

1. Withdrawn: The >10% Extension in Cycle Life

Resolution: withdrawal. J. Reviewer (battery safety officer) withdrew this hypothesis from the population in answer to a fatal flaw recorded by the safety and governance review. It did not enter the tournament, so it carries no rank and no Elo.

Fatal flaw recorded by the safety and governance review, verbatim:

> *Annealing PVDF-containing electrodes at 400°C will cause binder decomposition, releasing highly toxic and corrosive hydrogen fluoride (HF) gas, posing an extreme chemical safety hazard to researchers and equipment.*

Justification given by J. Reviewer (battery safety officer), verbatim:

> *Confirmed: PVDF decomposes above 350C and 400C annealing of an assembled electrode would vent HF into the lab. The confounder is worth testing, but only on binder-free electrodes, which is a different hypothesis.*

The >10% Extension in Cycle Life is therefore absent from the per-idea sections of this report as well. It was removed by decision rather than defeated on merit, and the population it was removed from is retained as superseded history.

Knowledge Base

Knowledge Summary

The following is the literature search's own summary of what it found. No source behind it was opened and checked in this run, so it is a report of what the search returned rather than a finding about the field, and the confidence in its wording is the search's own. The report evaluates the hypothesis that protective interphase coatings extend lithium-ion battery cycle life. Experimental data strongly supports that ultrathin inorganic and polymeric coatings suppress acid attack and structural pulverization, improving capacity retention. However, coatings exceeding optimal thickness or lacking ionic conductivity act as electrical bottlenecks, decreasing cycle life. The landscape highlights the necessity of rigorous testing constraints, uncoated control cells, and addresses scalability challenges and evidence gaps in cost modeling and long-term benchmarking.

Open Questions

The evidence that would change the recommendation is a list of its own, stated in full under Recommendations and Next Steps above; what follows is what the run left open besides it.

- There is a severe lack of granular, bottom-up cost models detailing the exact financial impact of integrating nanometer-precision coating processes into projected 2027-era gigafactories.
- There is a gap in standardized, multi-year degradation datasets specifically comparing coated versus uncoated cells across complex real-world operational profiles.
- There is missing data regarding the environmental impact and energy intensity of applying advanced chemical coatings in the Global South.
- A critical material disagreement centers on the trade-off between interfacial protection and ionic impedance, where coatings exceeding optimal thickness thresholds or lacking sufficient ionic conductivity act as electrical bottlenecks that actively decrease cycle life.
- The discovery pass found no adequate evidence under any facet it scores: supporting evidence, evidence contradicting the leading direction, negative or null results, independent replication, methodological detail, safety or governance evidence, and corrections or retractions affecting the sources used.
- The clustering stage fell back to a template, so what it recorded as coverage gaps is that template's default list rather than anything this run found missing, and it is not stated as an open question here.

Unexpected Connections

Nothing here was flagged as surprising; the run does not judge surprise. What this section reports is where ideas rest in the same cluster, which their separate rankings hide. What the clustering has not recorded is any mechanism that tells its clusters apart, as Main Research Directions above sets out, so a pair below shares a label and not a demonstrated failure mode: whether funding both buys two ideas' worth of information is left open here rather than answered. The remaining entry is the inverse case, a protected minority: one where a region rests on a single idea, so dropping it closes the region rather than narrowing it. It is about how thinly a region is covered rather than about any one idea's merits.

- Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating and A 15 nm Hybrid Organic-inorganic Coating Extends Cycle Life are the two ideas in the Evidence First cluster.

- A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode and A Self-assembling Amphiphilic Block-copolymer Coating on the Cathode are the two ideas in the Analogy Transfer cluster.
- A 2 nm Atomic Layer Deposition (ALD) Al2O3 Coating and A 10 nm Ductile PEDOT:PSS Conductive Polymer Coating are the two ideas in the Mechanism First cluster.
- A 5 nm Boron Nitride (BN) Coating was protected as a minority hypothesis. It ranks poorly on its own merits, but it is the only idea exploring its region of the problem, so discarding it would close off that region entirely. It is nonetheless outside any recommendation this report makes: a reviewer recorded a fatal flaw against it and the meta-review excluded it on that basis. Protection keeps the region open for a future run; it does not clear this idea to be acted on.

References

None of the sources below was checked against the document it names. They are the sources this report cites; what they are cited for is what the search said they hold, not what reading them established.

Where an entry names a publisher and nothing else, the literature search returned a redirect rather than the document, so it captured neither the document's title nor a link that reaches it.

1. Untitled source on mdpi.com. https://www.mdpi.com/2073-4360/18/4/429
2. Untitled source on electrochemsci.org. https://www.electrochemsci.org/papers/vol15/150908990.pdf
3. Untitled source on mdpi.com. https://www.mdpi.com/1996-1073/17/19/4970
4. Untitled source on researchgate.net. https://www.researchgate.net/publication/317195885_Tristrimethylsilyl_Phosphite_TMSPi_and_Triet…
5. Untitled source on nih.gov, the first of two separate records the search returned under that publisher without a title.
6. Untitled source on rsc.org.
7. Untitled source on nih.gov, the second of two separate records the search returned under that publisher without a title.
8. Untitled source on acs.org.
9. Untitled source on pcic.energy.

Top ideas in detail

Every idea below is set out the same way, so that a section can be compared across ideas rather than read only in place. What the fixed sections hold, and the qualifications that hold for every idea in this report, are stated here once rather than under each.

Description states the idea in the proposing specialist's own terms and in a fixed order: the mechanism it rests on, the predictions that separate it from its neighbours, the competing reading of the same situation, and the observation that would falsify it. A mechanism that does not hold takes everything under it with it; a prediction every competing idea also makes distinguishes nothing; and a falsifier stated before the work starts is what keeps the idea a hypothesis rather than a position.

Identified issues and validated risks are the risks the specialist that proposed the idea named against its own work. No reviewer was asked to confirm them, so the list is neither validated nor complete, and a risk missing from it has not been ruled out.

Addressed objections are the responses the reviews recorded. A review lists its objections and its responses separately without saying which answers which, so a response may concede an objection rather than dispose of it. Each response is attributed to the review that recorded it, so that is where a response is printed rather than a second time under the review itself, and a review that recorded none is not listed there rather than listed as silent. The objections themselves are under Deep Verification.

Supporting arguments state what the idea's mechanism predicts rather than the mechanism again, since a prediction specific enough to come out one way is what separates an idea worth running from one merely worth believing. Where findings are listed there, they are the ones the idea's own proposal cites; the evidence stage classified each of them for or against the research question and not against any one idea, so a finding can be listed under an idea it in fact tells against. No prediction anywhere in this report has been tested: the run proposes work rather than carrying any out, so a prediction is a case for the idea rather than a result of it. Goal alignment and novelty is a score and no more: what an idea has to displace is the competing reading set out under its own Description, and the score is a judgement about that contest rather than about the idea in isolation.

The feasibility assessment gives the review's score, what has to exist before the idea can be started at all, and the go/no-go tests. Those tests are what continuing or abandoning is decided against, and the specialist set them down before the work began, which is what keeps the decision from being made on the result once it is in hand.

The conclusion under each idea names the next move on it. Where an idea records both a go/no-go check and a falsifier, the go/no-go comes first: it is the cheaper of the two, and a failure there ends the work without the falsifier ever being run.

Where an idea carries a Revised Form section, only what the rewrite changed is set out in it: the fields that carry over unchanged are named there rather than printed a second time, and they are to be read from the idea above. Evolution rewrites the whole shortlist, so the heading reads Revised Form Recommended only where the meta-review went on to recommend the rewrite.

Where Critical Flaws reports that no reviewer recorded a fatal flaw, that is a statement about the reviews rather than a guarantee: a flaw nobody looked for is indistinguishable here from one that does not exist.

Deep Verification lists the fatal flaws and the objections the reviews raised, each numbered and attributed to the review that raised it, and all of them to be checked rather than argued away. Nothing in this run tested any item in those lists, so each is a live claim against the idea it is filed under rather than a settled one. A fatal flaw is the stronger finding: the reviewer that recorded one was saying the idea does not survive it, and the scoring scale caps such a review at two of five however confident it was. For the same reason no item there claims to have been answered: where a review responded at all, the responses are printed under Reviews, and which objection each one reaches is left to the reader to judge.

A debate appears under both of the ideas that fought it, each time argued from that section's side: the same exchange, read from the opposite end. The verdict and the rating change under it are the same in both places. A judge argues its verdict in the closing turn of the exchange, so where the verdict under a debate carries nothing further, its reasoning is that closing turn rather than missing; where the judge reasoned somewhere other than in the exchange, the verdict carries that reasoning, and where it recorded none the verdict says so.

Each idea below carries a grounding verdict under its title: four are marked uncited and three are marked unverified. What those verdicts mean is set out under Candidate Ideas above, where the ideas are first listed.

Every idea below is reviewed under the same headings, and each review asks the same question of every idea it reviews. Who asks what is set out here rather than under each idea; what varies, and what is printed there, is the answer.

- **Correctness —** Evidence and correctness reviewer. Is the claim correct, and is every load-bearing statement supported by evidence that has been inspected rather than merely retrieved?
- **Novelty** — Novelty reviewer. Does the idea go beyond established practice, and would a specialist in the field regard the contribution as new?
- **Feasibility** — Methods and statistics reviewer. Can the proposed design actually be executed, and would it yield an interpretable answer under realistic constraints?
- **Impact —** Impact reviewer. Would the result change what practitioners do, and is it worth the cost of finding out?
- **Safety** — Ethics, safety and governance reviewer. Does the idea raise a human-subject, biosafety, privacy, data-rights or dual-use concern, and which institutional approval must precede it?

Each idea below closes its reviews on whether they agree with each other. What a disagreement between them means, and what would settle one, is the same for every idea and is set out here; under each idea is that idea's own spread and the review at the bottom of it.

Where the reviews of an idea disagree by more than a point, the idea is strong on one dimension and weak on another rather than uniformly graded, and the review sections under it are to be read separately rather than averaged.

Where the lowest review of an idea is the evidence and correctness review, the disagreement is about the grounding rather than the design, and reading the cited sources in full is what would settle it. The falsifier would still decide the idea, but not until there is agreement on what the idea is built on.

Not every match was argued. Where an idea's table below shows a match with no debate under it, that match was decided by an arithmetic score comparison rather than by argued rounds: there is a verdict and the reason the judge recorded, and no exchange behind either to read.

One reason is recorded for every one of those matches, word for word, so it accounts for the tournament rather than for any single result: Compared evidence status, validity, novelty, feasibility, impact, risk, reproducibility, and information gain.

Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating

Rank: 1

Elo: 1262

Category: Experimental > Evidence First > Evidence-led

Evidence support: unverified.

Idea Proposal

Applying a 5 nm Copper-Tricarboxylate (CuTF) metal-organic framework coating on the electrode extends cycle life by >10% to 80% capacity retention compared to uncoated controls at 1C discharge rates.

Description

Evidence indicates that CuTF coatings can maintain 95% capacity after 280 cycles. An ultrathin MOF layer prevents transition metal crossover and acid attack while its porous structure maintains ionic pathways, avoiding the impedance trade-off seen in thicker coatings.

These are the predictions that separate it from its neighbours: coated cells will reach 80% capacity retention at least 10% later than uncoated controls, initial Coulombic efficiency of coated cells will be equal to or greater than controls, and internal resistance growth over 100 cycles will be significantly lower in coated cells.

Two competing readings of the same situation have to be displaced: the CuTF coating increases initial impedance, leading to lower initial capacity and failing the >15% capacity drop stopping criterion; and the coating degrades electrochemically at high voltages, providing no long-term benefit.

This is the result that would falsify it: cycle n=15 coated and n=15 uncoated control cells (randomized across cycler channels) at 1C within safe voltage limits (3.0-4.2V). Falsified if the mean cycles to 80% capacity retention for coated cells is not statistically significantly >10% higher than controls across 3 independent batches.

Revised Form

This is the form evolution produced, which the meta-review does not recommend, and it is not the form ranked above: evolution rewrote the idea in rounds one, two, and three, and this is the cumulative result. The rounds addressed, between them, unspecified effect size and external-validity assumptions. The rewrite leaves mechanism and rationale, alternative explanations, falsifier, required inputs and dependencies, principal risks, and go/no-go tests unchanged. It was re-reviewed after the rewrite: five reviews said revise it first.

Core idea: Applying a 5 nm Copper-Tricarboxylate (CuTF) metal-organic framework coating on the electrode extends cycle life by >10% to 80% capacity retention compared to uncoated controls at 1C discharge rates, under a preregistered discriminating design (revision 3).

Discriminating predictions: Coated cells will reach 80% capacity retention at least 10% later than uncoated controls, initial Coulombic efficiency of coated cells will be equal to or greater than controls, internal resistance growth over 100 cycles will be significantly lower in coated cells, and the primary result survives the prespecified robustness analysis, as prescribed by evolution round 3.

Reviews

Summary

1. Executive Verdict

This idea finished rank 1 on an Elo of 1262 and was carried forward onto the shortlist, averaging 4.4 of five across five reviews.

2. Critical Flaws

The correctness review recorded a fatal flaw against this idea. It is printed in full under Deep Verification below. Nothing in this run tested it, and no reviewer withdrew it.

3. Identified issues & Validated Risks

Coating process may be incompatible with standard cell assembly, and potential for thermal runaway if the MOF catalyzes electrolyte decomposition.

4. Addressed Objections

The correctness review answered: The idea could be revised to propose this as a pure hypothesis, but it currently states the performance as a known fact. The novelty review answered: The precise control of thickness to 5 nm to avoid the impedance trade-off provides specific engineering value and mechanistic insight. The feasibility review answered: Automated cycler data extraction inherently reduces observer bias, though analysis scripts should be pre-registered. The impact review answered: If performance gains are substantial, cost may be justified for premium applications. The safety review answered: The go/no-go tests include verifying initial capacity, which may indirectly indicate severe initial parasitic reactions, though direct gas monitoring would be safer.

5. Supporting Arguments & Evidence (Motivation)

No finding in this report's evidence is cited for this idea. The case for it is the discriminating prediction under Description.

6. Goal Alignment & Novelty

The novelty review scored this 5 of five, the highest any idea in this run received, shared with three other ideas.

7. Feasibility Assessment (Go/No-Go Decision)

The feasibility review scored this 5 of five. It cannot start until its inputs exist: ability to uniformly synthesize and apply 5 nm CuTF coatings, and calibrated battery cyclers with at least 30 available channels. Its go/no-go tests: verify initial capacity of coated cells is not >15% lower than uncoated controls, and confirm coating thickness is 5 nm ± 1 nm via TEM before full cell assembly.

8. Conclusion

The next move is the go/no-go work under Feasibility Assessment above, and the falsifier under Description after it. The lowest score it received is 2 of five, from the correctness review, printed in full below: that is the review to read before any of the above is commissioned.

Correctness

The idea relies on an unverified discovery lead (the source) as if it were verified evidence.

The claim about CuTF MOF coating maintaining 95% capacity after 280 cycles is based on an unverified source.

This review raised one objection and recorded one response.

Answer: evidence too thin to judge on, at the reviewer's stated confidence of 0.90

Score: 2

Novelty

Metal-organic frameworks (MOFs) like CuTF are gaining traction as battery coatings.

Testing a precisely 5 nm layer to balance transition metal blocking and ionic conductivity is a logical step from existing MOF literature.

This review raised one objection and recorded one response.

Answer: advance the idea as written, at the reviewer's stated confidence of 0.80

Score: 5

Feasibility

Includes intervention and control.

Specifies n=15 per group and 3 independent batches, satisfying replication requirements.

Includes randomization across cycler channels.

This review raised one objection and recorded one response.

Answer: advance the idea as written, at the reviewer's stated confidence of 0.90

Score: 5

Impact

Tests a novel MOF coating for selective ion transport.

High potential impact if successful, but scaling 5 nm uniform MOF coatings is challenging and costly.

This review raised one objection and recorded one response.

Answer: advance the idea as written, at the reviewer's stated confidence of 0.80

Score: 5

Safety

The idea involves testing MOF coatings which may catalyze electrolyte decomposition, posing a risk of thermal runaway.

Standard battery testing safety protocols, including BMS voltage limits and thermal chambers, are required.

This review raised one objection and recorded one response.

Answer: advance the idea as written, at the reviewer's stated confidence of 0.90

Score: 5

Coherence

The five reviews of this idea span 2 to 5 of five, a disagreement of more than a point. Part of that spread is a disqualification rather than a grade: the correctness review records a fatal flaw against the idea.

The lowest of them is the evidence and correctness review, at 2 of five: what it faults is the grounding, not the experiment.

Deep Verification

1. Fatal flaw recorded by the correctness review

Treats unverified discovery leads as verified evidence.

2. Raised by the correctness review

Without verified evidence, the specific performance claims of the CuTF coating cannot be substantiated.

3. Raised by the novelty review

MOFs for battery coatings are already a known and active research direction, making this somewhat incremental.

4. Raised by the feasibility review

No explicit blinding of the data analysis is mentioned.

5. Raised by the impact review

Cost of MOF precursors and deposition may be prohibitive for commercialization.

6. Raised by the safety review

If the MOF catalyzes rapid gas generation, standard venting mechanisms might be overwhelmed.

Tournament

Match summary

- Total matches: 4
- Matches won: 4
- Matches lost: 0
- Matches tied: 0
- Win rate: 100%

Round	Opponent	Result	Elo before	Elo after	Judge
1	A 5 nm Boron Nitride (BN) Coating	win	1200	1216	Arithmetic
2	A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode	win	1216	1232	Arithmetic
3	A 2 nm Atomic Layer Deposition (ALD) Al2O3 Coating	win	1232	1248	Arithmetic
4	A 15 nm Hybrid Organic-inorganic Coating Extends Cycle Life	win	1248	1262	Arithmetic

None of these matches carries a debate transcript.

A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode

Rank: 2

Elo: 1246

Category: Experimental > Analogy Transfer > Transfer-led

Evidence support: uncited.

Idea Proposal

A sacrificial lithium-carbonate rich composite pre-coating on the anode extends cycle life by >10% by preferentially reacting with trace water and HF to form a stable SEI, analogous to a sacrificial zinc anode in marine corrosion.

Description

In corrosion engineering, sacrificial materials protect the primary substrate. A reactive pre-coating can consume electrolyte impurities during the first formation cycle, creating a highly stable SEI that prevents long-term degradation of the uncoated baseline.

These are the predictions that separate it from its neighbours: coated cells will show >10% improvement in cycles to 80% capacity retention; first-cycle irreversible capacity loss may be higher for coated cells, but subsequent Coulombic efficiency will be superior; and gas generation during the first cycle will be higher for coated cells.

Two competing readings of the same situation have to be displaced: the sacrificial reaction generates excessive gas, causing cell rupture or delamination of the electrode; and the resulting SEI is highly resistive, leading to power fade.

This is the result that would falsify it: cycle n=15 coated and n=15 uncoated cells. Falsified if the cycle life to 80% retention does not improve by >10%, or if the initial capacity of coated cells is >15% lower than uncoated controls due to excessive sacrificial lithium consumption.

Revised Form

This is the form evolution produced, which the meta-review does not recommend, and it is not the form ranked above: evolution rewrote the idea in rounds one, two, and three, and this is the cumulative result. The rounds addressed, between them, unspecified effect size and external-validity assumptions. The rewrite leaves mechanism and rationale, alternative explanations, falsifier, required inputs and dependencies, principal risks, and go/no-go tests unchanged. It was re-reviewed after the rewrite: five reviews said revise it first.

Core idea: A sacrificial lithium-carbonate rich composite pre-coating on the anode extends cycle life by >10% by preferentially reacting with trace water and HF to form a stable SEI, analogous to a sacrificial zinc anode in marine corrosion, under a preregistered discriminating design (revision 3).

Discriminating predictions: Coated cells will show >10% improvement in cycles to 80% capacity retention; first-cycle irreversible capacity loss may be higher for coated cells, but subsequent Coulombic efficiency will be superior; gas generation during the first cycle will be higher for coated cells; and the primary result survives the prespecified robustness analysis, as prescribed by evolution round 3.

Reviews

Summary

1. Executive Verdict

This idea finished rank 2 on an Elo of 1246 and was carried forward onto the shortlist, averaging 3.4 of five across five reviews.

2. Critical Flaws

The correctness and feasibility reviews recorded two fatal flaws against this idea. They are printed in full under Deep Verification below. Nothing in this run tested them, and no reviewer withdrew any of them.

3. Identified issues & Validated Risks

Excessive gas generation during the sacrificial reaction could lead to cell venting or rupture, and handling highly reactive lithium composites poses fire safety risks.

4. Addressed Objections

The correctness review answered: While theoretically plausible, the lack of verified evidence makes it an unsupported proposal. The novelty review answered: Framing it specifically as a sacrificial impurity scavenger during the first cycle offers a distinct mechanistic angle compared to standard artificial SEIs. The feasibility review answered: Must revise to use specialized cells with gas release valves or rigid test cells with pressure monitoring, plus 3 batches. The impact review answered: Can be managed in a controlled dry room with specialized venting cells. The safety review answered: The proposal mentions a controlled environment (dry room), but explicit mention of Class D fire safety and institutional approval for reactive alkali metals is necessary.

5. Supporting Arguments & Evidence (Motivation)

No finding in this report's evidence is cited for this idea. The case for it is the discriminating prediction under Description.

6. Goal Alignment & Novelty

The novelty review scored this 5 of five, the highest any idea in this run received, shared with three other ideas.

7. Feasibility Assessment (Go/No-Go Decision)

The feasibility review scored this 2 of five. It cannot start until its inputs exist: controlled environment (dry room) for applying the moisture-sensitive sacrificial coating, and in-situ gas measurement capabilities (e.g., Archimedes method or DEMS). Its go/no-go tests: perform a single-cell formation cycle in a specialized containment vessel to quantify gas volume, and verify initial capacity drop is <15% compared to controls.

8. Conclusion

The next move is the go/no-go work under Feasibility Assessment above, and the falsifier under Description after it. Its lowest score, 2 of five, is shared by the correctness and feasibility reviews, both printed in full below: they are what to read before any of the above is commissioned.

Correctness

The idea proposes an analogy-based mechanism (sacrificial lithium-carbonate pre-coating) without citing any verified evidence.

This review raised one objection and recorded one response.

Answer: evidence too thin to judge on, at the reviewer's stated confidence of 0.90

Score: 2

Novelty

Using a sacrificial lithium-carbonate composite to pre-form SEI and consume impurities is an interesting translation of the sacrificial anode concept from corrosion engineering.

This review raised one objection and recorded one response.

Answer: advance the idea as written, at the reviewer's stated confidence of 0.80

Score: 5

Feasibility

Specifies n=15 per group.

This review raised one objection and recorded one response.

Answer: revise it first, at the reviewer's stated confidence of 0.90, with the score capped by a fatal flaw

Score: 2

Impact

Innovative approach using a sacrificial layer.

Handling reactive lithium-carbonate composites poses significant safety and manufacturing challenges.

High risk of excessive gas generation during formation.

This review raised two objections and recorded one response.

Answer: revise it first, at the reviewer's stated confidence of 0.80

Score: 4

Safety

The idea involves handling highly reactive lithium composites, which poses severe fire safety risks.

Excessive gas generation during the sacrificial reaction could lead to cell venting or rupture.

This review raised one objection and recorded one response.

Answer: revise it first, at the reviewer's stated confidence of 0.95

Score: 4

Coherence

The five reviews of this idea span 2 to 5 of five, a disagreement of more than a point. Part of that spread is a disqualification rather than a grade: the correctness and feasibility reviews record two fatal flaws against the idea.

The lowest of them is the evidence and correctness review, at 2 of five: what it faults is the grounding, not the experiment.

Deep Verification

1. Fatal flaw recorded by the correctness review

Lacks verified evidence to support the proposed mechanism and specific parameters.

2. Fatal flaw recorded by the feasibility review

Fails to specify replication across 3 independent batches.

3. Raised by the correctness review

The specific expected outcomes (>10% improvement) are speculative without evidence.

4. Raised by the novelty review

Pre-lithiation and artificial SEIs are known concepts, which overlaps somewhat with this proposal.

5. Raised by the feasibility review

If gas generation is high, pouch cells will swell, causing loss of contact and impedance rise, which confounds the cycle life measurement.

6. Raised by the impact review

Safety risks of handling reactive Li composites in standard environments.

7. Raised by the impact review

Gas generation may cause cell rupture.

8. Raised by the safety review

Handling bare reactive lithium composites requires strict institutional safety approvals and specialized fire suppression (Class D) that are not explicitly mandated in the proposal.

Tournament

Match summary

- Total matches: 5
- Matches won: 4
- Matches lost: 1
- Matches tied: 0
- Win rate: 80%

Round	Opponent	Result	Elo before	Elo after	Judge
1	A Self-assembling Amphiphilic Block-copolymer Coating on the Cathode	win	1200	1216	Arithmetic
2	Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating	loss	1216	1200	Arithmetic
3	A 5 nm Boron Nitride (BN) Coating	win	1200	1216	Arithmetic
4	A 2 nm Atomic Layer Deposition (ALD) Al2O3 Coating	win	1216	1232	Arithmetic
4	A 15 nm Hybrid Organic-inorganic Coating Extends Cycle Life	win	1232	1246	Arithmetic

None of these matches carries a debate transcript.

A 2 nm Atomic Layer Deposition (ALD) Al2O3 Coating

Rank: 3

Elo: 1199

Category: Experimental > Mechanism First > Mechanism-led

Evidence support: unverified.

Idea Proposal

A 2 nm Atomic Layer Deposition (ALD) Al_2O_3 coating on NCM cathodes extends cycle life by >10% by chemically scavenging HF generated from $LiPF_6$ degradation, preventing transition metal dissolution.

Description

Cycle life degradation in high-nickel cathodes is primarily driven by HF attack from electrolyte decomposition. Al_2O_3 reacts with HF to form stable AlF_3, protecting the active material without blocking Li^+ transport due to its ultrathin nature.

These are the predictions that separate it from its neighbours: coated cells will demonstrate >10% improvement in cycles to 80% capacity retention; post-mortem analysis of the electrolyte will show significantly lower concentrations of dissolved Ni, Co, and Mn in coated cells; and initial Coulombic efficiency will be maintained or improved.

Two competing readings of the same situation have to be displaced: the Al_2O_3 coating acts purely as a physical barrier rather than a chemical scavenger, and the 2 nm coating is too thin to provide meaningful HF scavenging over hundreds of cycles.

This is the result that would falsify it: cycle n=15 coated and n=15 uncoated cells across 3 independent batches. Measure dissolved transition metals in the electrolyte post-mortem via ICP-MS. Falsified if cycle life is not extended by >10% OR if transition metal dissolution is not significantly reduced in coated cells.

Revised Form

This is the form evolution produced, which the meta-review does not recommend, and it is not the form ranked above: evolution rewrote the idea in rounds one, two, and three, and this is the cumulative result. The rounds addressed, between them, unspecified effect size and external-validity assumptions. The rewrite leaves mechanism and rationale, alternative explanations, falsifier, required inputs and dependencies, principal risks, and go/no-go tests unchanged. It was re-reviewed after the rewrite: five reviews said revise it first.

Core idea: A 2 nm Atomic Layer Deposition (ALD) Al_2O_3 coating on NCM cathodes extends cycle life by >10% by chemically scavenging HF generated from $LiPF_6$ degradation, preventing transition metal dissolution, under a preregistered discriminating design (revision 3).

Discriminating predictions: Coated cells will demonstrate >10% improvement in cycles to 80% capacity retention; post-mortem analysis of the electrolyte will show significantly lower concentrations of dissolved Ni, Co, and Mn in coated cells; initial Coulombic efficiency will be maintained or improved; and the primary result survives the prespecified robustness analysis, as prescribed by evolution round 3.

Reviews

Summary

1. Executive Verdict

This idea finished rank 3 on an Elo of 1199 and was carried forward onto the shortlist, averaging 3.4 of five across five reviews.

2. Critical Flaws

The correctness and novelty reviews recorded two fatal flaws against this idea. They are printed in full under Deep Verification below. Nothing in this run tested them, and no reviewer withdrew any of them.

3. Identified issues & Validated Risks

ALD precursors may react with the cathode surface, altering baseline performance; and handling of HF-containing degraded electrolytes poses chemical safety risks.

4. Addressed Objections

Four reviews of this idea recorded a response. The correctness review answered: The idea could be revised to propose this as a pure hypothesis, but it currently states the performance as a known fact. The feasibility review answered: The protocol can be revised to include acid digestion and ICP-MS of the harvested anodes alongside the electrolyte. The impact review answered: Spatial ALD is being developed for continuous roll-to-roll processing. The safety review answered: Standard battery lab protocols generally cover electrolyte handling, but HF requires specific antidotes (e.g., calcium gluconate) and training.

5. Supporting Arguments & Evidence (Motivation)

No finding in this report's evidence is cited for this idea. The case for it is the discriminating prediction under Description.

6. Goal Alignment & Novelty

The novelty review scored this 2 of five, the lowest any idea in this run received, shared with two other ideas.

7. Feasibility Assessment (Go/No-Go Decision)

The feasibility review scored this 4 of five. It cannot start until its inputs exist: access to ALD equipment capable of conformal 2 nm deposition, and ICP-MS availability for post-mortem transition metal quantification. Its go/no-go tests: confirm conformal 2 nm Al_2O_3 coverage via HR-TEM, and verify initial capacity of coated cells is not >15% lower than uncoated controls.

8. Conclusion

The next move is the go/no-go work under Feasibility Assessment above, and the falsifier under Description after it. Its lowest score, 2 of five, is shared by the correctness and novelty reviews, both printed in full below: they are what to read before any of the above is commissioned.

Correctness

The idea relies on an unverified discovery lead (the source) as if it were verified evidence.

The claim about a 2 nm ALD Al_2O_3 coating scavenging HF is based on an unverified source.

This review raised one objection and recorded one response.

Answer: evidence too thin to judge on, at the reviewer's stated confidence of 0.90

Score: 2

Novelty

ALD Al_2O_3 on NCM cathodes is one of the most heavily researched topics in battery coatings over the last decade.

This review raised one objection.

Answer: reject it, at the reviewer's stated confidence of 0.95

Score: 2

Feasibility

Includes intervention and control.

Specifies n=15 per group across 3 batches.

Uses ICP-MS for dependent variable measurement.

This review raised one objection and recorded one response.

Answer: revise it first, at the reviewer's stated confidence of 0.90

Score: 4

Impact

Directly tests the HF scavenging mechanism of ALD Al_2O_3.

High information gain regarding chemical vs physical protection.

ALD is expensive but provides precise thickness control.

This review raised one objection and recorded one response.

Answer: advance the idea as written, at the reviewer's stated confidence of 0.90

Score: 5

Safety

The idea requires post-mortem analysis of electrolytes to measure dissolved transition metals.

Degraded $LiPF_6$ electrolytes contain hydrofluoric acid (HF), which is highly toxic and corrosive.

This review raised one objection and recorded one response.

Answer: revise it first, at the reviewer's stated confidence of 0.95

Score: 4

Coherence

The five reviews of this idea span 2 to 5 of five, a disagreement of more than a point. Part of that spread is a disqualification rather than a grade: the correctness and novelty reviews record two fatal flaws against the idea.

One review raised one objection and recorded no response to it — the Novelty review. An objection nobody answered is not a refuted one, and it still applies to the idea as written.

The lowest of them is the evidence and correctness review, at 2 of five: what it faults is the grounding, not the experiment.

Deep Verification

1. Fatal flaw recorded by the correctness review

Treats unverified discovery leads as verified evidence.

2. Fatal flaw recorded by the novelty review

Lacks novelty entirely. The mechanism of HF scavenging by Al_2O_3 to prevent transition metal dissolution is already widely accepted and proven in prior art.

3. Raised by the correctness review

Without verified evidence, the specific performance claims of the Al2O3 coating cannot be substantiated.

4. Raised by the novelty review, unanswered

This is a textbook example of an incremental replication study rather than a novel hypothesis.

5. Raised by the feasibility review

Transition metals often deposit on the anode; measuring only the electrolyte will systematically underestimate dissolution and increase measurement uncertainty.

6. Raised by the impact review

ALD is too slow and expensive for gigafactory scale.

7. Raised by the safety review

The proposal does not explicitly mandate HF-specific safety protocols or institutional chemical safety approvals for the post-mortem extraction.

Tournament

Match summary

- Total matches: 4
- Matches won: 2
- Matches lost: 2
- Matches tied: 0
- Win rate: 50%

Round	Opponent	Result	Elo before	Elo after	Judge
1	A 15 nm Hybrid Organic-inorganic Coating Extends Cycle Life	win	1200	1216	Arithmetic
2	A 10 nm Ductile PEDOT:PSS Conductive Polymer Coating	win	1216	1231	Arithmetic
3	Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating	loss	1231	1215	Arithmetic
4	A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode	loss	1215	1199	Arithmetic

None of these matches carries a debate transcript.

A 5 nm Boron Nitride (BN) Coating

Rank: 4

Elo: 1184

Category: Experimental > Competing Explanation > Falsification-led

Evidence support: uncited.

Idea Proposal

A 5 nm Boron Nitride (BN) coating extends cycle life by >10% primarily by acting as a thermal heat spreader that eliminates localized hot spots on the electrode surface, rather than by acting as a chemical barrier against the electrolyte.

Description

Localized Joule heating accelerates SEI degradation and transition metal dissolution. A thermally conductive but electrically insulating coating homogenizes the temperature profile, slowing down these thermally activated degradation mechanisms.

These are the predictions that separate it from its neighbours: BN-coated cells will show >10% improvement in cycles to 80% capacity retention, in-situ thermal imaging will show a significantly more uniform temperature distribution across the BN-coated electrode compared to controls, and cells coated with a thermally insulating material of the same thickness will not show the cycle life improvement.

Two competing readings of the same situation have to be displaced: the BN coating improves cycle life purely through chemical passivation, independent of its thermal conductivity; and localized hot spots are not a primary driver of degradation under standardized cycling conditions.

This is the result that would falsify it: cycle n=15 BN-coated cells, n=15 uncoated cells, and n=15 cells coated with a thermally insulating polymer of the same thickness. Falsified if BN-coated cells do not outperform both other groups by >10% in cycle life, or if thermal imaging shows no reduction in localized hot spots.

Revised Form

This is the form evolution produced, which the meta-review does not recommend, and it is not the form ranked above: evolution rewrote the idea in rounds one, two, and three, and this is the cumulative result. The rounds addressed, between them, unspecified effect size and external-validity assumptions. The rewrite leaves mechanism and rationale, alternative explanations, falsifier, required inputs and dependencies, principal risks, and go/no-go tests unchanged. It was re-reviewed after the rewrite: five reviews said revise it first.

Core idea: A 5 nm Boron Nitride (BN) coating extends cycle life by >10% primarily by acting as a thermal heat spreader that eliminates localized hot spots on the electrode surface, rather than by acting as a chemical barrier against the electrolyte, under a preregistered discriminating design (revision 3).

Discriminating predictions: BN-coated cells will show >10% improvement in cycles to 80% capacity retention, in-situ thermal imaging will show a significantly more uniform temperature distribution across the BN-coated electrode compared to controls, cells coated with a thermally insulating material of the same thickness will not show the cycle life improvement, and the primary result survives the prespecified robustness analysis, as prescribed by evolution round 3.

Reviews

Summary

1. Executive Verdict

This idea finished rank 4 on an Elo of 1184 and was carried forward onto the shortlist, averaging 3.4 of five across five reviews.

2. Critical Flaws

The correctness and feasibility reviews recorded two fatal flaws against this idea. They are printed in full under Deep Verification below. Nothing in this run tested them, and no reviewer withdrew any of them.

3. Identified issues & Validated Risks

Modified cell designs for thermal imaging may not accurately represent standard commercial cell behavior, and BN deposition processes may introduce impurities that cause internal short circuits.

4. Addressed Objections

The correctness review answered: While theoretically plausible, the lack of verified evidence makes it an unsupported proposal. The novelty review answered: The novelty lies in the experimental design (comparing with a thermally insulating coating) to isolate the thermal vs. chemical mechanism, which is rarely done. The feasibility review answered: Revise to use internal micro-thermocouples in standard cell packaging instead of IR windows, and add 3 batches. The impact review answered: Can be supplemented with post-mortem analysis of standard cells. The safety review answered: This risk can be mitigated by requiring hydrostatic pressure testing of the modified cell casings prior to introducing active materials and electrolytes.

5. Supporting Arguments & Evidence (Motivation)

No finding in this report's evidence is cited for this idea. The case for it is the discriminating prediction under Description.

6. Goal Alignment & Novelty

The novelty review scored this 5 of five, the highest any idea in this run received, shared with three other ideas.

7. Feasibility Assessment (Go/No-Go Decision)

The feasibility review scored this 2 of five. It cannot start until its inputs exist: capability to deposit uniform 5 nm BN coatings and a thermally insulating control coating, and specialized cell designs with IR-transparent windows for in-situ thermal imaging. Its go/no-go tests: verify the initial capacity of BN-coated cells is not >15% lower than uncoated controls, and confirm the thermal conductivity of the BN coating is at least an order of magnitude higher than the insulating control coating.

8. Conclusion

The next move is the go/no-go work under Feasibility Assessment above, and the falsifier under Description after it. Its lowest score, 2 of five, is shared by the correctness and feasibility reviews, both printed in full below: they are what to read before any of the above is commissioned.

Correctness

The idea proposes a competing explanation (5 nm Boron Nitride coating acting as a thermal heat spreader) without citing any verified evidence.

This review raised one objection and recorded one response.

Answer: evidence too thin to judge on, at the reviewer's stated confidence of 0.90

Score: 2

Novelty

Testing Boron Nitride (BN) specifically for its thermal heat-spreading properties to eliminate localized hot spots, rather than its chemical passivation, is a novel mechanistic decoupling.

This review raised one objection and recorded one response.

Answer: advance the idea as written, at the reviewer's stated confidence of 0.85

Score: 5

Feasibility

3-arm design (BN, uncoated, insulating).

Specifies n=15 per group.

Proposes in-situ thermal imaging.

This review raised one objection and recorded one response.

Answer: revise it first, at the reviewer's stated confidence of 0.90, with the score capped by a fatal flaw

Score: 2

Impact

Tests an interesting hypothesis about thermal vs chemical protection.

In-situ thermal imaging requires specialized cell designs (e.g., window cells) that alter thermal mass and boundary conditions, reducing external validity.

This review raised one objection and recorded one response.

Answer: revise it first, at the reviewer's stated confidence of 0.85

Score: 4

Safety

The idea requires specialized cell designs with IR-transparent windows for in-situ thermal imaging.

Modified cell casings may compromise the structural integrity and pressure containment of the cell.

This review raised one objection and recorded one response.

Answer: revise it first, at the reviewer's stated confidence of 0.95

Score: 4

Coherence

The five reviews of this idea span 2 to 5 of five, a disagreement of more than a point. Part of that spread is a disqualification rather than a grade: the correctness and feasibility reviews record two fatal flaws against the idea.

The lowest of them is the evidence and correctness review, at 2 of five: what it faults is the grounding, not the experiment.

Deep Verification

1. Fatal flaw recorded by the correctness review

Lacks verified evidence to support the proposed mechanism and specific parameters.

2. Fatal flaw recorded by the feasibility review

Fails to specify replication across 3 independent batches.

3. Raised by the correctness review

The specific expected outcomes (>10% improvement) are speculative without evidence.

4. Raised by the novelty review

BN is already used as a coating, so the material itself is not novel.

5. Raised by the feasibility review

The IR window alters heat dissipation, confounding the measurement of the BN coating's thermal effect and external validity.

6. Raised by the impact review

Window cells have completely different thermal dissipation properties than standard cylindrical or pouch cells, invalidating the thermal comparison.

7. Raised by the safety review

If the IR window fails under pressure, it could result in a sudden release of flammable electrolyte and toxic gases, or projectile hazards.

Tournament

Match summary

- Total matches: 3
- Matches won: 1
- Matches lost: 2
- Matches tied: 0
- Win rate: 33%

Round	Opponent	Result	Elo before	Elo after	Judge
1	Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating	loss	1200	1184	Arithmetic
2	A Self-assembling Amphiphilic Block-copolymer Coating on the Cathode	win	1184	1200	Arithmetic
3	A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode	loss	1200	1184	Arithmetic

None of these matches carries a debate transcript.

A 15 nm Hybrid Organic-inorganic Coating Extends Cycle Life

Rank: 5

Elo: 1172

Category: Experimental > Evidence First > Evidence-led

Evidence support: unverified.

Idea Proposal

A 15 nm hybrid organic-inorganic coating extends cycle life by >10% at low discharge rates (0.2C) but decreases cycle life compared to uncoated controls at high rates (2.0C) due to restricted Li+ permeability.

Description

Evidence shows hybrid coatings reduce permeability, increasing DC internal resistance at high rates (>1.0C) while showing no capacity difference at low rates. This suggests a rate-dependent impedance trade-off where the coating acts as an electrical bottleneck under high current.

These are the predictions that separate it from its neighbours: at 0.2C, coated cells will show >10% improvement in cycles to 80% capacity retention vs controls; at 2.0C, coated cells will reach 80% capacity retention faster than uncoated controls; and DC internal resistance will be significantly higher for coated cells at 2.0C.

Two competing readings of the same situation have to be displaced: the coating provides universal protection, extending cycle life at both 0.2C and 2.0C; and the coating is too thick and paralyzes ion transport completely, failing at all C-rates.

This is the result that would falsify it: test n=15 coated and n=15 uncoated cells at 0.2C, and another set of n=15 coated and n=15 uncoated cells at 2.0C. Falsified if coated cells do not show >10% life extension at 0.2C OR if they do not show worse capacity retention than controls at 2.0C.

Reviews

Summary

1. Executive Verdict

This idea finished rank 5 on an Elo of 1172 and was held back from the shortlist, averaging 3.2 of five across five reviews.

2. Critical Flaws

The correctness, novelty, and feasibility reviews recorded three fatal flaws against this idea. They are printed in full under Deep Verification below. Nothing in this run tested them, and no reviewer withdrew any of them.

3. Identified issues & Validated Risks

High-rate cycling of high-impedance cells may lead to localized overheating and thermal runaway, and power fade may trigger cycler safety voltage limits prematurely.

4. Addressed Objections

The correctness review answered: The idea could be revised to propose this as a pure hypothesis, but it currently states the performance as a known fact. The novelty review answered: Quantifying the exact C-rate threshold could be useful, though it does not constitute a novel scientific discovery. The feasibility review answered: Can be revised to include 3 batches and adjust the high-rate C-rate if safety cutoffs are triggered. The impact review answered: The conceptual framework of rate-dependent impedance is broadly applicable. The safety review answered: The proposal mandates verifying BMS safety cutoffs are active for the 2.0C testing group, which should mitigate catastrophic failure.

5. Supporting Arguments & Evidence (Motivation)

No finding in this report's evidence is cited for this idea. The case for it is the discriminating prediction under Description.

6. Goal Alignment & Novelty

The novelty review scored this 2 of five, the lowest any idea in this run received, shared with two other ideas.

7. Feasibility Assessment (Go/No-Go Decision)

The feasibility review scored this 2 of five. It cannot start until its inputs exist: precise control over hybrid coating thickness to exactly 15 nm, and environmental chambers capable of maintaining strict temperature control during high-rate cycling. Its go/no-go tests: conduct Electrochemical Impedance Spectroscopy (EIS) to ensure initial resistance is within safe testing limits, and verify BMS safety cutoffs are active for the 2.0C testing group.

8. Conclusion

It is not on the shortlist, so nothing is scheduled against it. If it is revived, the first move is the go/no-go work under Feasibility Assessment above, and the falsifier under Description after it. Its lowest score, 2 of five, is shared by the correctness, novelty, and feasibility reviews, all printed in full below: they are what to read before any of the above is commissioned.

Correctness

The idea relies on an unverified discovery lead (the source) as if it were verified evidence.

The claim about a 15 nm hybrid coating decreasing cycle life at high rates is based on an unverified source.

This review raised one objection and recorded one response.

Answer: evidence too thin to judge on, at the reviewer's stated confidence of 0.90

Score: 2

Novelty

Hybrid organic-inorganic coatings are extensively documented in prior art.

Investigating rate-dependent impedance and the trade-off between coating thickness and high-rate performance is a well-known phenomenon.

This review raised one objection and recorded one response.

Answer: reject it, at the reviewer's stated confidence of 0.90

Score: 2

Feasibility

2x2 factorial design (coated/uncoated x 0.2C/2.0C).

Specifies n=15 per group.

This review raised two objections and recorded one response.

Answer: revise it first, at the reviewer's stated confidence of 0.90, with the score capped by a fatal flaw

Score: 2

Impact

Addresses the critical trade-off between protection and impedance at high C-rates.

Highly relevant for fast-charging applications.

Feasible to test with standard equipment.

This review raised one objection and recorded one response.

Answer: advance the idea as written, at the reviewer's stated confidence of 0.90

Score: 5

Safety

High-rate (2.0C) cycling of cells with potentially high-impedance coatings poses a significant risk of localized overheating and thermal runaway.

The proposal correctly identifies power fade and premature safety voltage limits as risks.

This review raised one objection and recorded one response.

Answer: advance the idea as written, at the reviewer's stated confidence of 0.90

Score: 5

Coherence

The five reviews of this idea span 2 to 5 of five, a disagreement of more than a point. Part of that spread is a disqualification rather than a grade: the correctness, novelty, and feasibility reviews record three fatal flaws against the idea.

The lowest of them is the evidence and correctness review, at 2 of five: what it faults is the grounding, not the experiment.

Deep Verification

1. Fatal flaw recorded by the correctness review

Treats unverified discovery leads as verified evidence.

2. Fatal flaw recorded by the novelty review

Highly incremental; the battery community already widely acknowledges that thick coatings restrict high-rate performance and act as electrical bottlenecks.

3. Fatal flaw recorded by the feasibility review

Fails to specify replication across 3 independent batches as required by the success criteria.

4. Raised by the correctness review

Without verified evidence, the specific performance claims of the hybrid coating cannot be substantiated.

5. Raised by the novelty review

The core hypothesis merely replicates established knowledge regarding ion transport paralysis in thick coatings.

6. Raised by the feasibility review

Missing replication across batches.

7. Raised by the feasibility review

High-rate cycling of high-impedance cells may trigger safety cutoffs before true capacity fade is measured.

8. Raised by the impact review

Results may be highly specific to the chosen hybrid material.

9. Raised by the safety review

High-rate testing of restricted-permeability cells could lead to rapid lithium plating and subsequent internal short circuits.

Tournament

Match summary

- Total matches: 4
- Matches won: 1

- Matches lost: 3
- Matches tied: 0
- Win rate: 25%

Round	Opponent	Result	Elo before	Elo after	Judge
1	A 2 nm Atomic Layer Deposition (ALD) Al2O3 Coating	loss	1200	1184	Arithmetic
3	A 10 nm Ductile PEDOT:PSS Conductive Polymer Coating	win	1184	1200	Arithmetic
4	Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating	loss	1200	1186	Arithmetic
4	A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode	loss	1186	1172	Arithmetic

None of these matches carries a debate transcript.

A 10 nm Ductile PEDOT:PSS Conductive Polymer Coating

Rank: 6

Elo: 1169

Category: Experimental > Mechanism First > Mechanism-led

Evidence support: uncited.

Idea Proposal

A 10 nm ductile PEDOT:PSS conductive polymer coating on silicon-graphite anodes extends cycle life by >10% by accommodating volume expansion and preventing mechanical pulverization of the SEI.

Description

Cycle life degradation in high-capacity anodes is driven by mechanical stress and continuous SEI fracture during lithiation/delithiation. A ductile, conductive coating maintains electrical contact and stabilizes the interphase during volume changes.

These are the predictions that separate it from its neighbours: coated cells will achieve >10% improvement in cycles to 80% capacity retention, cross-sectional SEM will show significantly less particle cracking and SEI thickness growth in coated cells compared to controls, and coulombic efficiency will stabilize faster in coated cells.

Two competing readings of the same situation have to be displaced: the polymer coating degrades electrochemically at low anode potentials, and the coating restricts lithium diffusion, causing lithium plating and rapid capacity fade.

This is the result that would falsify it: cycle n=15 coated and n=15 uncoated cells. Falsified if coated cells do not achieve >10% improvement in cycles to 80% capacity retention, or if post-mortem cross-sectional SEM shows equal or greater particle pulverization compared to controls.

Reviews

Summary

1. Executive Verdict

This idea finished rank 6 on an Elo of 1169 and was held back from the shortlist, averaging 3.2 of five across five reviews.

2. Critical Flaws

The correctness, novelty, and feasibility reviews recorded three fatal flaws against this idea. They are printed in full under Deep Verification below. Nothing in this run tested them, and no reviewer withdrew any of them.

3. Identified issues & Validated Risks

Residual water in the polymer coating could react with the electrolyte, causing severe gas generation and cell venting; and lithium plating risk if the coating impedes ion transport.

4. Addressed Objections

The correctness review answered: While theoretically plausible, the lack of verified evidence makes it an unsupported proposal. The novelty review answered: Specific thickness optimization might yield marginal gains, but the core concept is not novel. The feasibility review answered: Can be revised to include 3 batches and quantitative image analysis over multiple random cross-sections, or combined with macroscopic thickness measurements. The impact review answered: Strict Karl Fischer titration constraints mitigate this risk. The safety review answered: The proposed go/no-go test of ensuring moisture content is below 50 ppm proactively addresses the root cause of the severe gas generation risk.

5. Supporting Arguments & Evidence (Motivation)

No finding in this report's evidence is cited for this idea. The case for it is the discriminating prediction under Description.

6. Goal Alignment & Novelty

The novelty review scored this 2 of five, the lowest any idea in this run received, shared with two other ideas.

7. Feasibility Assessment (Go/No-Go Decision)

The feasibility review scored this 2 of five. It cannot start until its inputs exist: synthesis of battery-grade, moisture-free PEDOT:PSS; and cross-sectional SEM capabilities for post-mortem analysis. Its go/no-go tests: Karl Fischer titration to ensure moisture content of the coated electrode is below 50 ppm, and verify no lithium plating occurs during the first 5 formation cycles.

8. Conclusion

It is not on the shortlist, so nothing is scheduled against it. If it is revived, the first move is the go/no-go work under Feasibility Assessment above, and the falsifier under Description after it. Its lowest score, 2 of five, is shared by the correctness, novelty, and feasibility reviews, all printed in full below: they are what to read before any of the above is commissioned.

Correctness

The idea proposes a specific mechanism (10 nm ductile PEDOT:PSS coating) without citing any verified evidence.

This review raised one objection and recorded one response.

Answer: evidence too thin to judge on, at the reviewer's stated confidence of 0.90

Score: 2

Novelty

PEDOT:PSS and other conductive polymer coatings on silicon anodes to buffer volume expansion are well-documented in existing literature.

This review raised one objection and recorded one response.

Answer: reject it, at the reviewer's stated confidence of 0.90

Score: 2

Feasibility

Specifies n=15 per group.

Proposes cross-sectional SEM for post-mortem analysis.

This review raised one objection and recorded one response.

Answer: revise it first, at the reviewer's stated confidence of 0.90, with the score capped by a fatal flaw

Score: 2

Impact

Targets the mechanical degradation of Si anodes, a major industry bottleneck.

PEDOT:PSS is commercially available and relatively cheap.

High external validity for next-generation high-energy cells.

This review raised one objection and recorded one response.

Answer: advance the idea as written, at the reviewer's stated confidence of 0.85

Score: 5

Safety

Residual water in the PEDOT:PSS coating could react with the electrolyte, causing severe gas generation and cell venting.

Lithium plating is identified as a risk if the coating impedes ion transport.

This review raised one objection and recorded one response.

Answer: advance the idea as written, at the reviewer's stated confidence of 0.90

Score: 5

Coherence

The five reviews of this idea span 2 to 5 of five, a disagreement of more than a point. Part of that spread is a disqualification rather than a grade: the correctness, novelty, and feasibility reviews record three fatal flaws against the idea.

The lowest of them is the evidence and correctness review, at 2 of five: what it faults is the grounding, not the experiment.

Deep Verification

1. Fatal flaw recorded by the correctness review

Lacks verified evidence to support the proposed mechanism and specific parameters.

2. Fatal flaw recorded by the novelty review

The use of ductile conductive polymers to accommodate Si anode volume expansion is a highly saturated research area.

3. Fatal flaw recorded by the feasibility review

Fails to specify replication across 3 independent batches.

4. Raised by the correctness review

The specific parameters (10 nm) and expected outcomes (>10% improvement) are speculative without evidence.

5. Raised by the novelty review

Both the material (PEDOT:PSS) and the mechanism (mechanical buffering of SEI) are standard in current battery research.

6. Raised by the feasibility review

SEM is qualitative and prone to sampling bias; it lacks a quantitative metric for particle cracking across a statistically significant area.

7. Raised by the impact review

Residual moisture in PEDOT:PSS is notoriously difficult to remove completely and will react with LiPF6.

8. Raised by the safety review

If venting occurs, the release of toxic and flammable gases (e.g., HF, organic carbonates) must be managed by appropriate lab exhaust systems.

Tournament

Match summary

- Total matches: 2
- Matches won: 0
- Matches lost: 2
- Matches tied: 0
- Win rate: 0%

Round	Opponent	Result	Elo before	Elo after	Judge
2	A 2 nm Atomic Layer Deposition (ALD) Al2O3 Coating	loss	1200	1185	Arithmetic
3	A 15 nm Hybrid Organic-inorganic Coating Extends Cycle Life	loss	1185	1169	Arithmetic

None of these matches carries a debate transcript.

A Self-assembling Amphiphilic Block-copolymer Coating on the Cathode

Rank: 7

Elo: 1168

Category: Experimental > Analogy Transfer > Transfer-led

Evidence support: uncited.

Idea Proposal

A self-assembling amphiphilic block-copolymer coating on the cathode extends cycle life by >10% by acting as a selective filter that allows bare Li+ transport while sterically hindering the co-intercalation of large solvent molecules, analogous to biological ion channels.

Description

Biological cell membranes use size and charge exclusion to achieve selective permeability. An engineered polymer with specific pore sizes can prevent solvent molecules (e.g., EC, DEC) from reaching the active material surface, halting parasitic side reactions.

These are the predictions that separate it from its neighbours: coated cells will show >10% improvement in cycles to 80% capacity retention, electrolyte consumption over 200 cycles will be significantly lower in coated cells, and initial Coulombic efficiency will be improved due to reduced solvent oxidation.

Two competing readings of the same situation have to be displaced: the polymer pores become blocked by degradation products, paralyzing ion transport; and the polymer dissolves in the organic carbonate electrolyte over time.

This is the result that would falsify it: cycle n=15 coated and n=15 uncoated cells. Falsified if coated cells fail to show >10% cycle life improvement, or if impedance spectroscopy shows the coating completely paralyzes Li+ transport resulting in an initial capacity >15% lower than controls.

Reviews

Summary

1. Executive Verdict

This idea finished rank 7 on an Elo of 1168 and was held back from the shortlist, averaging 3.6 of five across five reviews.

2. Critical Flaws

The correctness and feasibility reviews recorded three fatal flaws against this idea. They are printed in full under Deep Verification below. Nothing in this run tested them, and no reviewer withdrew any of them.

3. Identified issues & Validated Risks

Polymer dissolution could increase electrolyte viscosity and decrease overall cell safety, and complete pore blockage could lead to rapid lithium plating and short circuits.

4. Addressed Objections

The correctness review answered: While theoretically plausible, the lack of verified evidence makes it an unsupported proposal. The novelty review answered: As a fundamental experimental study, it provides high information gain regarding selective permeability in interphases, pushing the boundary of coating design. The feasibility review answered: Revise to increase sample size to allow for destructive teardowns at intervals (e.g., n=45 per group per batch) and include 3 batches. The impact review answered: Cross-linking the polymer post-deposition could enhance stability. The safety review answered: The go/no-go test to verify initial capacity and impedance should catch completely blocked pores before long-term cycling and dendrite growth occur.

5. Supporting Arguments & Evidence (Motivation)

No finding in this report's evidence is cited for this idea. The case for it is the discriminating prediction under Description.

6. Goal Alignment & Novelty

The novelty review scored this 5 of five, the highest any idea in this run received, shared with three other ideas.

7. Feasibility Assessment (Go/No-Go Decision)

The feasibility review scored this 2 of five. It cannot start until its inputs exist: synthesis of block-copolymers with precise sub-nanometer pore sizes, and verification of coating insolubility in standard carbonate electrolytes. Its go/no-go tests: soak coated electrodes in electrolyte at 45°C for 7 days to verify coating insolubility, and verify initial capacity of coated cells is not >15% lower than uncoated controls.

8. Conclusion

It is not on the shortlist, so nothing is scheduled against it. If it is revived, the first move is the go/no-go work under Feasibility Assessment above, and the falsifier under Description after it. Its lowest score, 2 of five, is shared by the correctness and feasibility reviews, both printed in full below: they are what to read before any of the above is commissioned.

Correctness

The idea proposes an analogy-based mechanism (self-assembling amphiphilic block-copolymer coating) without citing any verified evidence.

This review raised one objection and recorded one response.

Answer: evidence too thin to judge on, at the reviewer's stated confidence of 0.90

Score: 2

Novelty

Designing a block-copolymer as a selective ion channel to sterically hinder solvent co-intercalation is a highly novel, biomimetic approach.

This review raised one objection and recorded one response.

Answer: advance the idea as written, at the reviewer's stated confidence of 0.85

Score: 5

Feasibility

Specifies n=15 per group.

This review raised one objection and recorded one response.

Answer: revise it first, at the reviewer's stated confidence of 0.90, with the score capped by a fatal flaw

Score: 2

Impact

Biomimetic approach to selective ion transport.

Synthesizing block-copolymers with precise sub-nanometer pores that are insoluble in carbonate electrolytes is extremely difficult.

High risk of polymer dissolution or pore blockage.

This review raised one objection and recorded one response.

Answer: revise it first, at the reviewer's stated confidence of 0.85

Score: 4

Safety

Complete pore blockage of the polymer coating could lead to rapid lithium plating and internal short circuits.

Polymer dissolution could increase electrolyte viscosity and decrease overall cell safety.

This review raised one objection and recorded one response.

Answer: advance the idea as written, at the reviewer's stated confidence of 0.90

Score: 5

Coherence

The five reviews of this idea span 2 to 5 of five, a disagreement of more than a point. Part of that spread is a disqualification rather than a grade: the correctness and feasibility reviews record three fatal flaws against the idea.

The lowest of them is the evidence and correctness review, at 2 of five: what it faults is the grounding, not the experiment.

Deep Verification

1. Fatal flaw recorded by the correctness review

Lacks verified evidence to support the proposed mechanism and specific parameters.

2. Fatal flaw recorded by the feasibility review

Fails to specify replication across 3 independent batches.

3. Fatal flaw recorded by the feasibility review

Sample size is insufficient for the proposed dependent variable (electrolyte consumption over 200 cycles).

4. Raised by the correctness review

The specific expected outcomes (>10% improvement) are speculative without evidence.

5. Raised by the novelty review

Synthesizing such precise polymers may be too complex for practical battery applications.

6. Raised by the feasibility review

Measuring electrolyte consumption requires destructive teardowns. A sample size of n=15 is too small if cells must be sacrificed at multiple intervals to plot consumption over time.

7. Raised by the impact review

Carbonate electrolytes are strong solvents that typically swell or dissolve such polymers.

8. Raised by the safety review

Internal short circuits from lithium dendrites can bypass external safety circuits if they occur rapidly.

Tournament

Match summary

- Total matches: 2
- Matches won: 0
- Matches lost: 2
- Matches tied: 0
- Win rate: 0%

Round	Opponent	Result	Elo before	Elo after	Judge
1	A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode	loss	1200	1184	Arithmetic
2	A 5 nm Boron Nitride (BN) Coating	loss	1184	1168	Arithmetic

None of these matches carries a debate transcript.

Provenance

Run

- Session: session_6305ce9003fb4658 (the id this run's saved session, ledger rows and stored artifacts are filed under)
- Started: 2026-08-04 06:46:47 UTC
- Last recorded activity: 2026-08-04 07:37:07 UTC, 50 minutes after it started
- Workflow version: 2
- Stages completed: 8 of 8
- Approvals: every stage gate in this run was accepted automatically under the auto approval profile, so acceptance here records a well-formed payload rather than a person's agreement — the warning under Research Goal above says what it does not amount to
- Produced by: a fixed template (not a model), deep-research-preview-04-2026, and gemini-3.1-pro-preview
- Prompt version: 1
- Tournament protocol configured: 3 Swiss rounds then a top-4 round robin
- Tournament as played: 4 rounds of 3, 3, 3, and 3 matches, 12 in all; its last round played 3 matches between 4 ideas, where a round robin over them is 6 matches
- Tournament outcome: stopped without converging; no rating change was recorded for its final round
- Judged by: an arithmetic score comparison

Literature discovery

Deep Research ran one pass, at an estimated cost of $3.00, and returned 53 source leads. It stopped because the pass limit was reached with coverage still short of sufficient.

Evidence integrity

Every idea in this run carries a qualification on its grounding: no evidence was cited for the idea at all, or the evidence it cites is absent from this session, or it was retracted, or it was retrieved but never checked against its source. Each line states one of the four and names the ideas it applies to.

- Three ideas rest on evidence that was discovered but never verified, so each of their claims is a hypothesis: Applying a 5 nm Copper-Tricarboxylate (CuTF) Metal-organic Framework Coating; A 2 nm Atomic Layer Deposition (ALD) Al2O3 Coating; and A 15 nm Hybrid Organic-inorganic Coating Extends Cycle Life.
- Four ideas cite no evidence at all. The specialists that proposed them recorded no source, so nothing in this session grounds them either way and each claim is a conjecture: A Sacrificial Lithium-carbonate Rich Composite Pre-coating on the Anode; A 5 nm Boron Nitride (BN) Coating; A 10 nm Ductile PEDOT:PSS Conductive Polymer Coating; and A Self-assembling Amphiphilic Block-copolymer Coating on the Cathode.

What each stage produced

Stage	Specialist	What it produced	Produced by	Written by
scope	goal scoping	research plan	gemini-3.1-pro-preview	specialist
evidence	deep research discovery	literature discovery manifest	deep-research-preview-04-2026	specialist
evidence	literature discovery	evidence packet	gemini-3.1-pro-preview	specialist
evidence	source verification	evidence packet	gemini-3.1-pro-preview	specialist
generate	idea generation	hypothesis population	gemini-3.1-pro-preview	specialist
reflect	evidence and correctness review	set of reviews	gemini-3.1-pro-preview	specialist
reflect	novelty review	set of reviews	gemini-3.1-pro-preview	specialist
reflect	methods and statistics review	set of reviews	gemini-3.1-pro-preview	specialist
reflect	ethics, safety and governance review	set of reviews	gemini-3.1-pro-preview	specialist
reflect	impact review	set of reviews	gemini-3.1-pro-preview	specialist
rank	tournament ranking	tournament record	a fixed template (not a model)	deterministic fallback
evolve	evolution of the shortlist	revision cycle	a fixed template (not a model)	deterministic fallback
proximity	clustering by mechanism	idea landscape	a fixed template (not a model)	deterministic fallback
meta-review	meta-review	meta-review of the round	a fixed template (not a model)	deterministic fallback

The candidate population was rewritten once after a governance adjudication withdrew a hypothesis. Only the current version is listed above; the earlier version is retained in the session as superseded rather than deleted, so the population each stage actually ran against stays readable.

Four stages fell back to a fixed template because the specialist's answer came back incomplete or malformed. The affected sections carry an inline warning; on a healthy run no stage falls back at all.